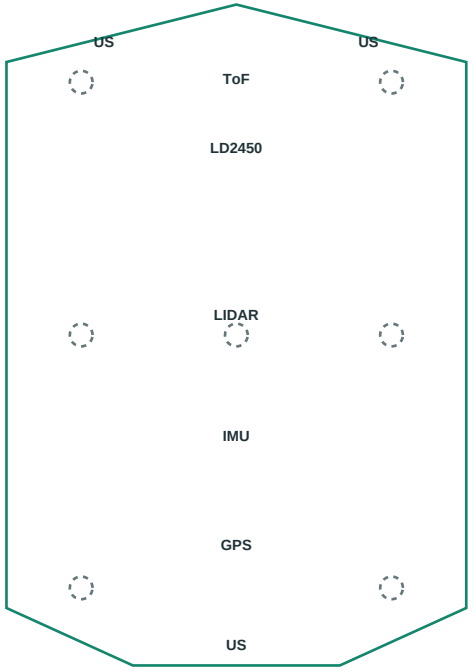
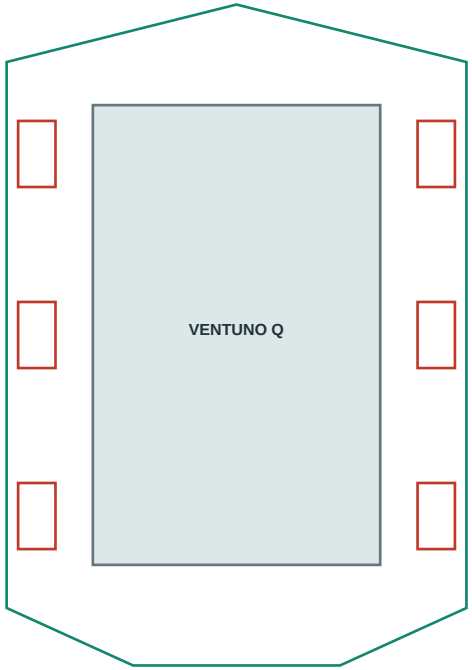
CARDBOARD MOCKUP - SENSOR CHASSIS

A4 print pack for a quick physical fit check of the VENTUNO Q sensor carrier. Templates are full size unless marked schematic.

Exploded arrangement (schematic, not to scale)



UPPER DECK / sensors face outward



LOWER DECK / board and servo lands

Side stack

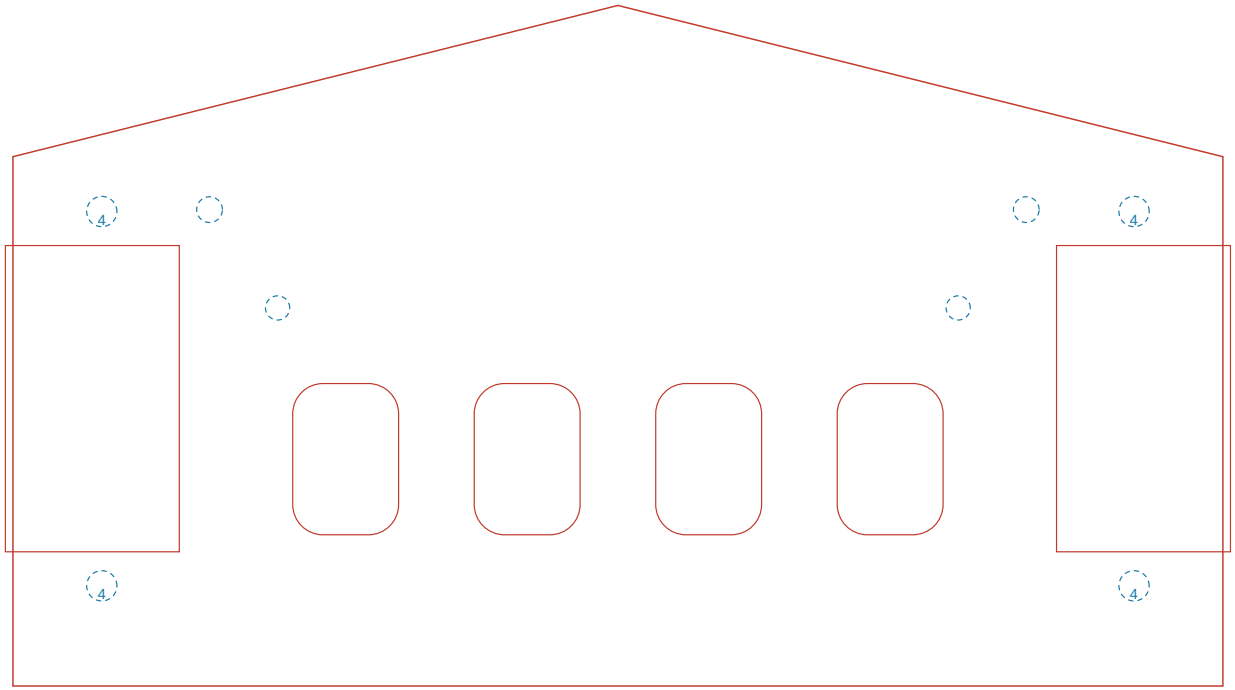
Print and assemble

1. Print pages 2-11 at Actual size, landscape A4.
2. Measure the 50 mm check box before cutting.
3. Cut six deck panels; tape the seams from below.
4. Use splice marks and sensor footprints as guides.
5. Fold/roll the holder nets on pages 8-10.
6. Make six 55 mm spacer tubes from page 11.
7. Mockup only: cardboard is not load-bearing.

Warning: board-hole coordinates and sensor fit are provisional. Confirm against the actual modules before cutting the final cardboard pieces.

LOWER DECK - FRONT SECTION

Full scale 1:1 template. Cut the red outline and openings; pierce the blue fastener marks.



CUT / ASSEMBLY

Red: cut line / opening

Blue dashed: drill/pierce

Green: splice strap holes

Envelope: 160 x 90 mm

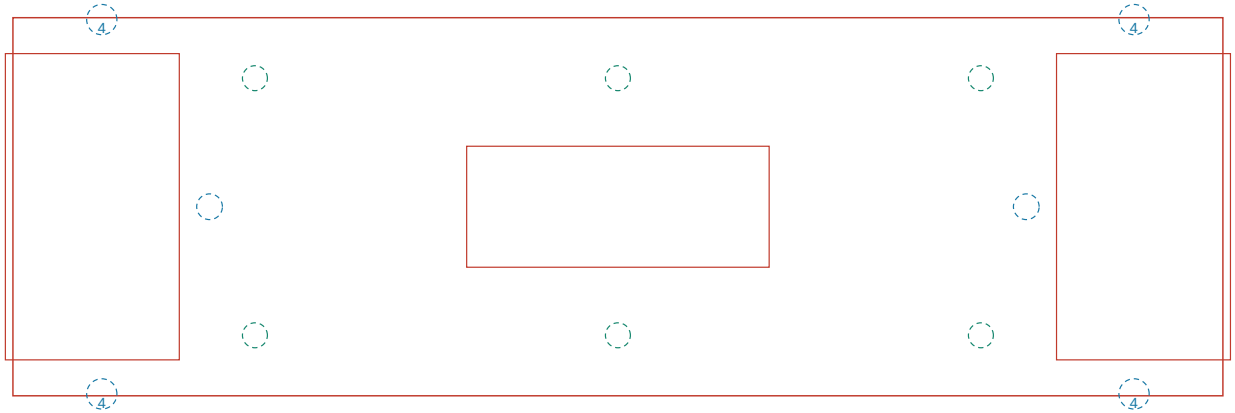
Part thickness: 3.5 mm

Holes near a seam may form an edge notch; use the straps to bridge the joint underneath.

50 mm CHECK

LOWER DECK - CENTER SECTION

Full scale 1:1 template. Cut the red outline and openings; pierce the blue fastener marks.



CUT / ASSEMBLY

Red: cut line / opening

Blue dashed: drill/pierce

Green: splice strap holes

Envelope: 160 x 50 mm

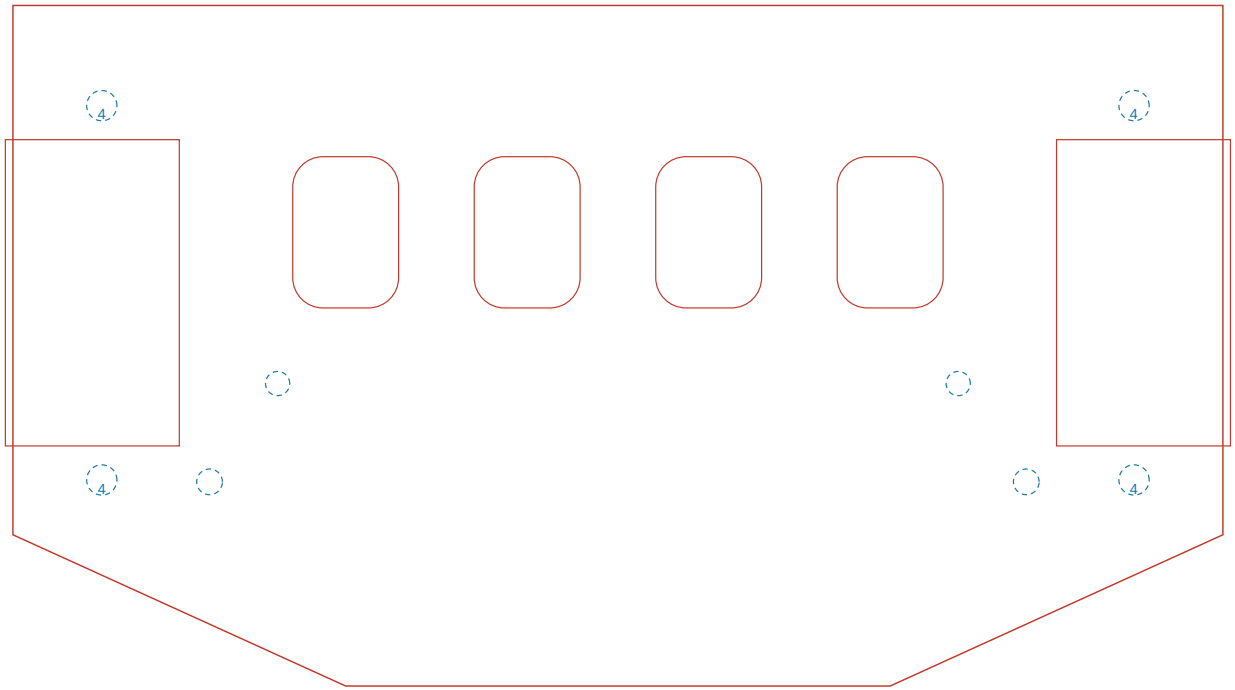
Part thickness: 3.5 mm

Holes near a seam may form an edge notch; use the straps to bridge the joint underneath.

50 mm CHECK

LOWER DECK - REAR SECTION

Full scale 1:1 template. Cut the red outline and openings; pierce the blue fastener marks.



CUT / ASSEMBLY

Red: cut line / opening

Blue dashed: drill/pierce

Green: splice strap holes

Envelope: 160 x 90 mm

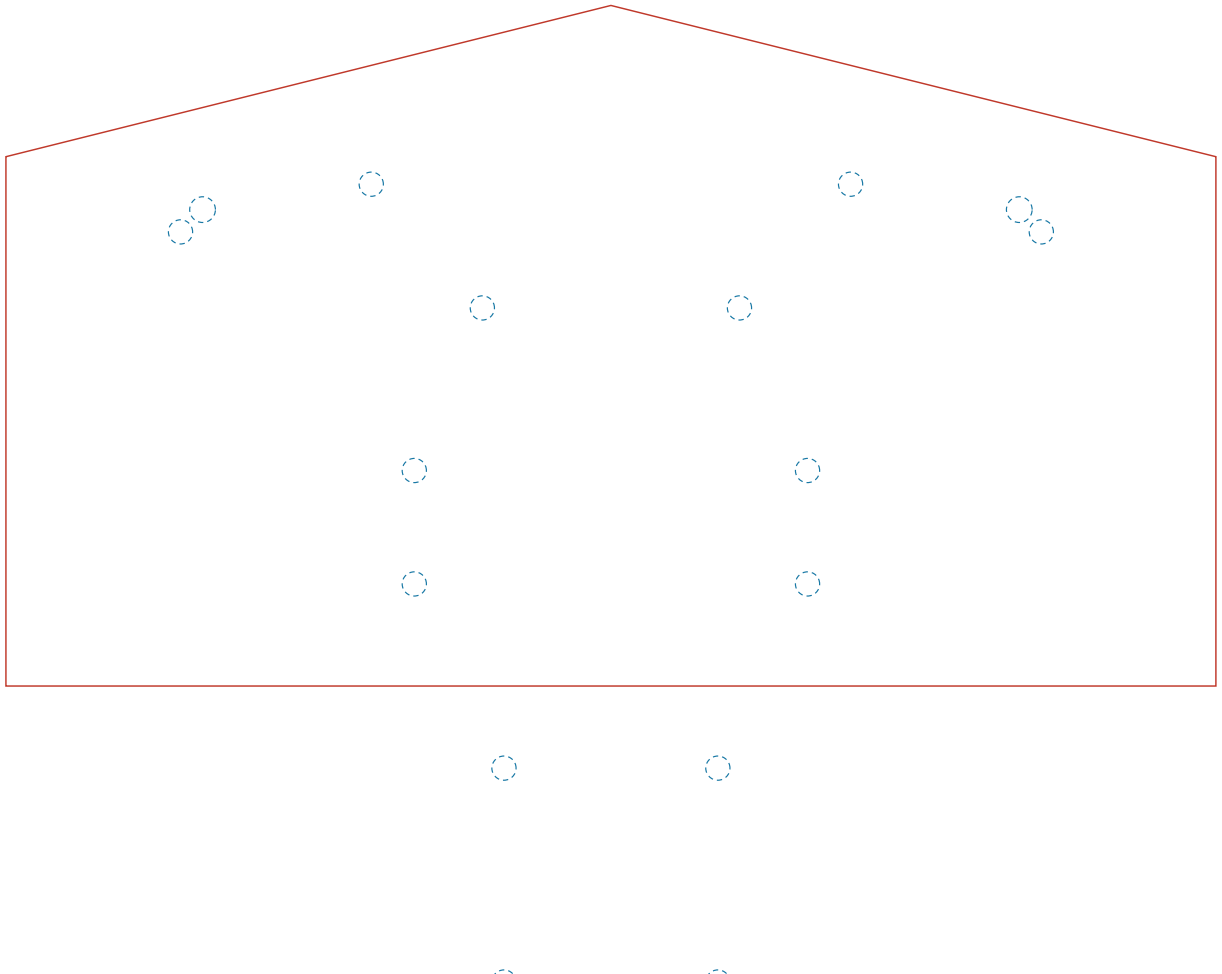
Part thickness: 3.5 mm

Holes near a seam may form an edge notch; use the straps to bridge the joint underneath.

50 mm CHECK

UPPER DECK - FRONT SECTION

Full scale 1:1 template. Cut the red outline and openings; pierce the blue fastener marks.



CUT / ASSEMBLY

Red: cut line / opening

Blue dashed: drill/pierce

Green: splice strap holes

Envelope: 160 x 90 mm

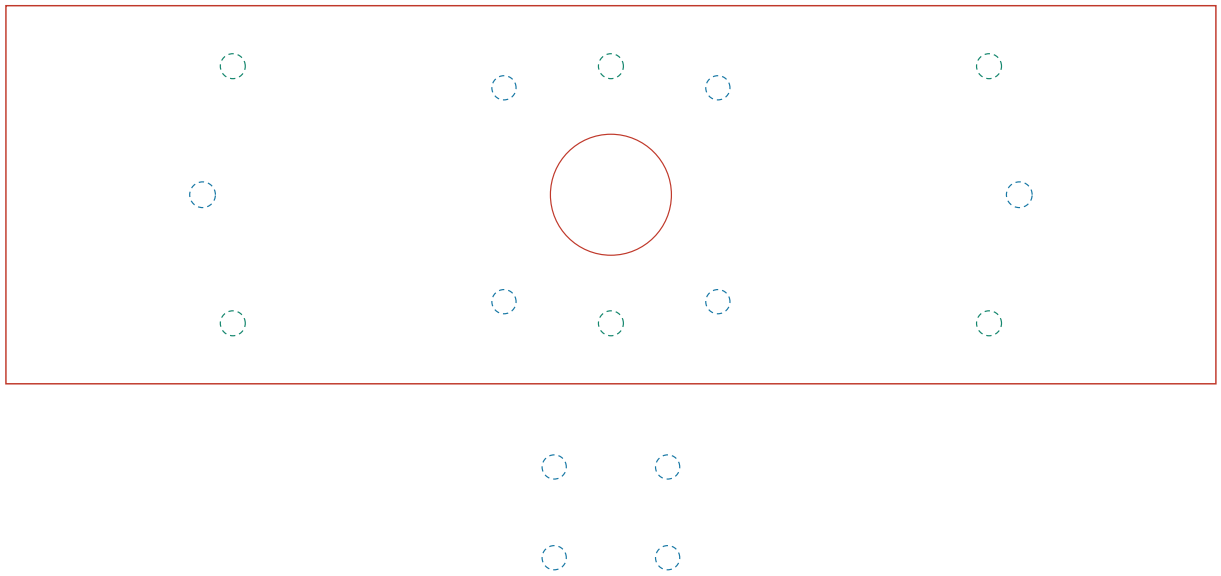
Part thickness: 3.5 mm

Holes near a seam may form an edge notch; use the straps to bridge the joint underneath.

50 mm CHECK

UPPER DECK - CENTER SECTION

Full scale 1:1 template. Cut the red outline and openings; pierce the blue fastener marks.



CUT / ASSEMBLY

Red: cut line / opening

Blue dashed: drill/pierce

Green: splice strap holes

Envelope: 160 x 50 mm

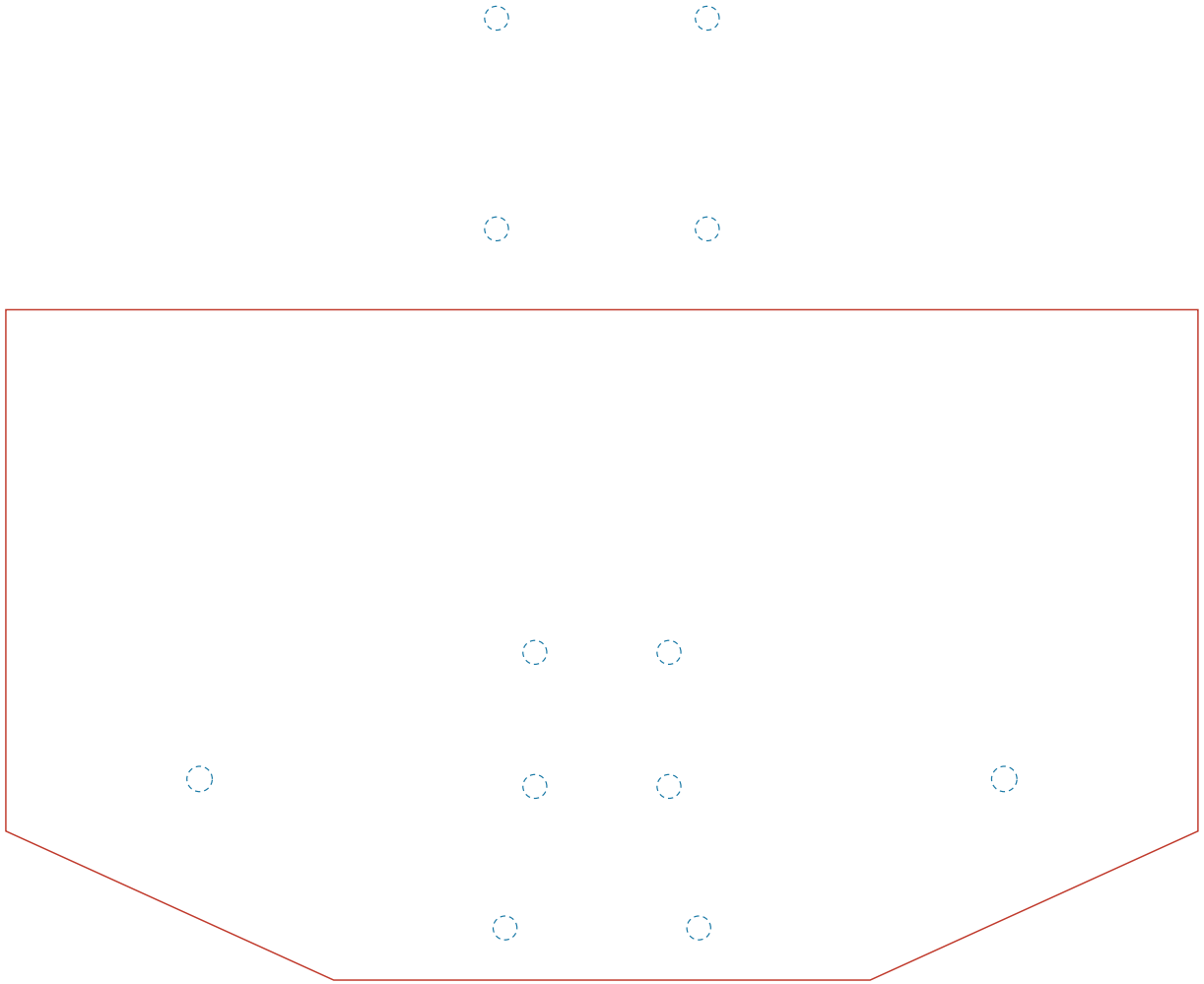
Part thickness: 3.5 mm

Holes near a seam may form an edge notch; use the straps to bridge the joint underneath.

50 mm CHECK

UPPER DECK - REAR SECTION

Full scale 1:1 template. Cut the red outline and openings; pierce the blue fastener marks.



CUT / ASSEMBLY

Red: cut line / opening

Blue dashed: drill/pierce

Green: splice strap holes

Envelope: 160 x 90 mm

Part thickness: 3.5 mm

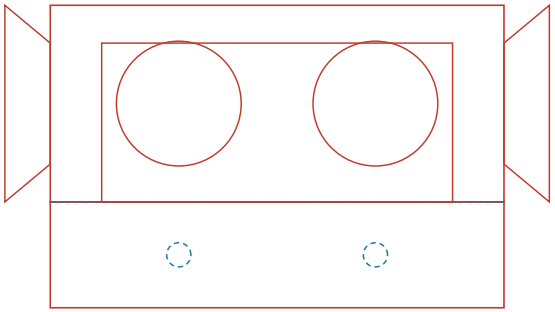
Holes near a seam may form an edge notch; use the straps to bridge the joint underneath.

50 mm CHECK

CARDBOARD HOLDERS - ULTRASONIC + RADAR

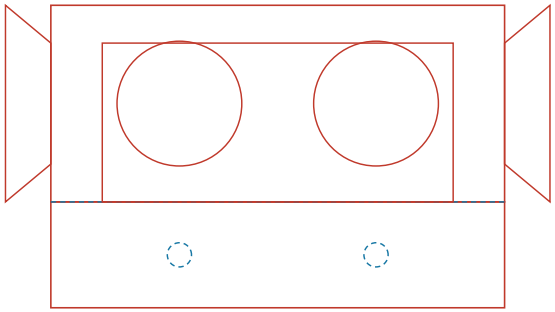
Fold lines are blue dashed. Red is cut. Approximate envelope/template from the current STL geometry.

HC-SR04 cradle - make 3



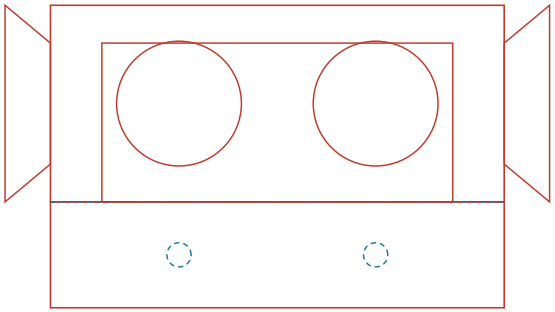
Two 16.5 mm transducer holes; scale 1:1.

HC-SR04 cradle - make 3



Two 16.5 mm transducer holes; scale 1:1.

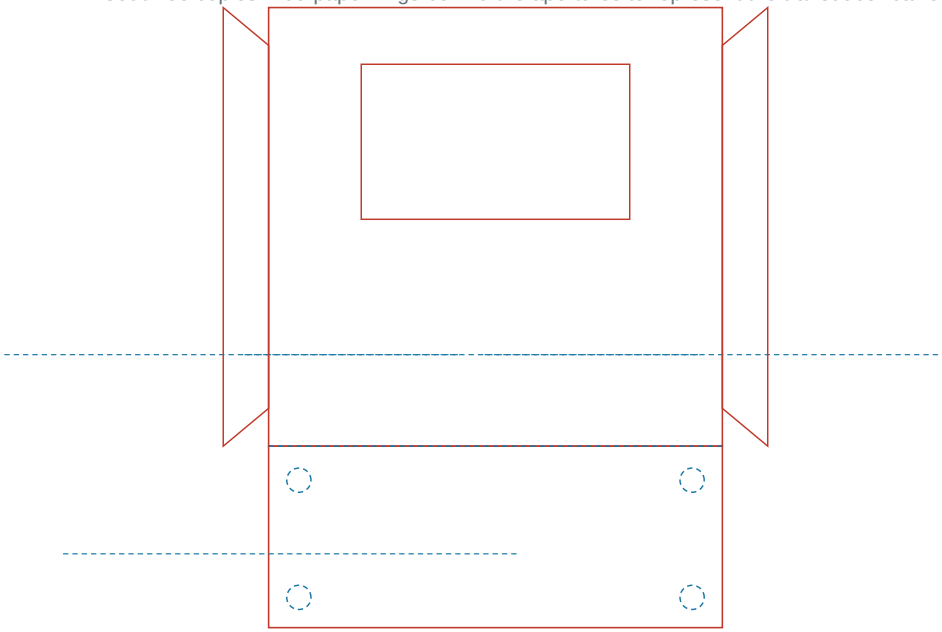
HC-SR04 cradle - make 3



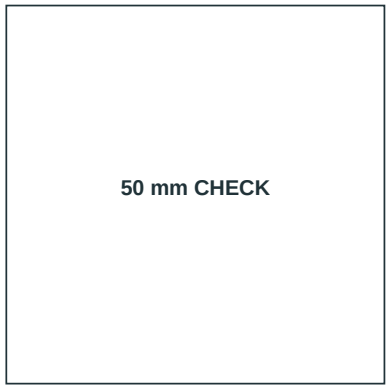
Two 16.5 mm transducer holes; scale 1:1.

LD2450 radar tower base + wall

Cut three copies. Add paper rings behind the apertures to represent the transducer barrels.



Wall window 35.5 x 20.5 mm; leave it open for the mockup.



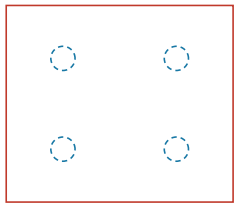
Cardboard holder nets approximate the STL shape; use tape at the tabs. Do not attach powered electronics to cardboard.

CARDBOARD HOLDERS - ToF / IMU / DECK JOINTS

Fold the ToF side walls upward; stack IMU pads; tape splice straps under the deck seams.

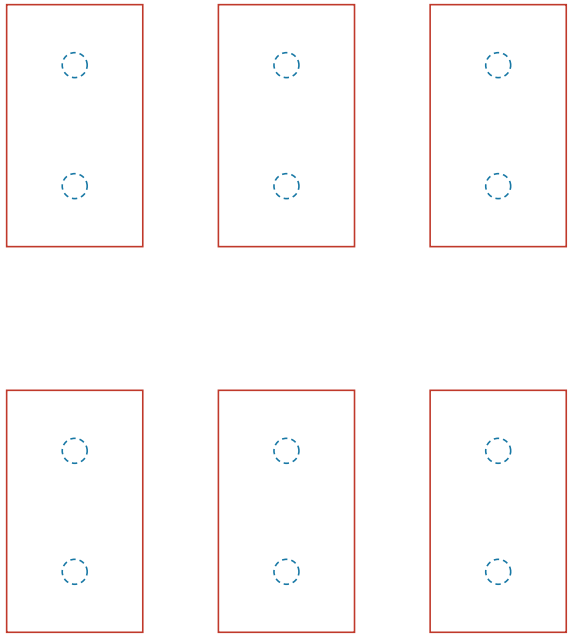
2 side ears 23 x 4; rear lip 36 x 3. Glue upright at marked base edges.

IMU pad - stack 1 or 2 layers



Glue/fold narrow rails into a U cradle. Leave the sensor lens end clear.

Deck splice strap 18 x 32 mm - cut 12



Tape 3 straps under each seam. M3 hardware belongs to the later rigid build.

50 mm CHECK

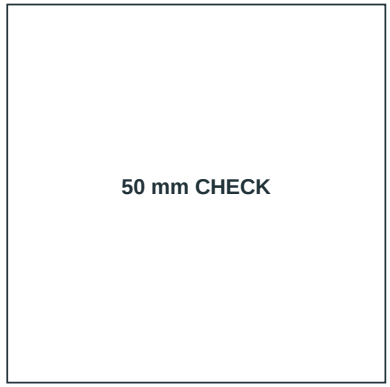
Red solid = cut; blue dashed = pierce/fold. All templates on this page are 1:1.

CARDBOARD HOLDERS - MASTS + SPACERS

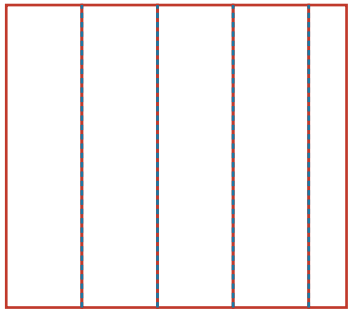
Cut, roll, and tape paper forms to check heights and collisions. Dimensions are 1:1 except notes marked schematic.

LIDAR mast - roll 34 mm diameter x 75 mm high

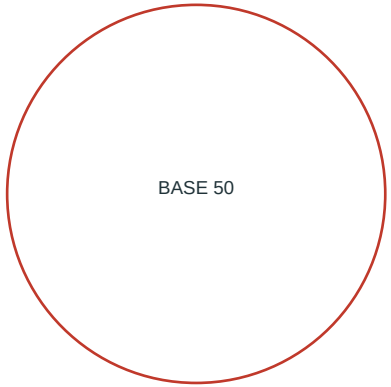
Strip includes 8 mm glue flap; base circle dia 50 mm, top plate dia 72 mm.

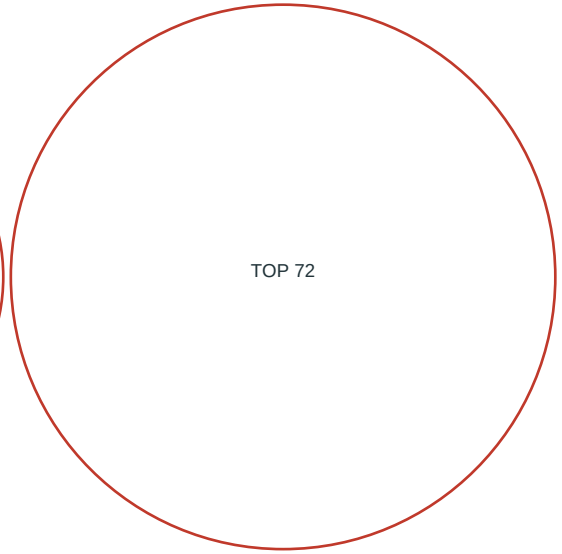


GPS mast wrap: 10 x 10 x 40 mm



Fold into square tube; glue the edge.



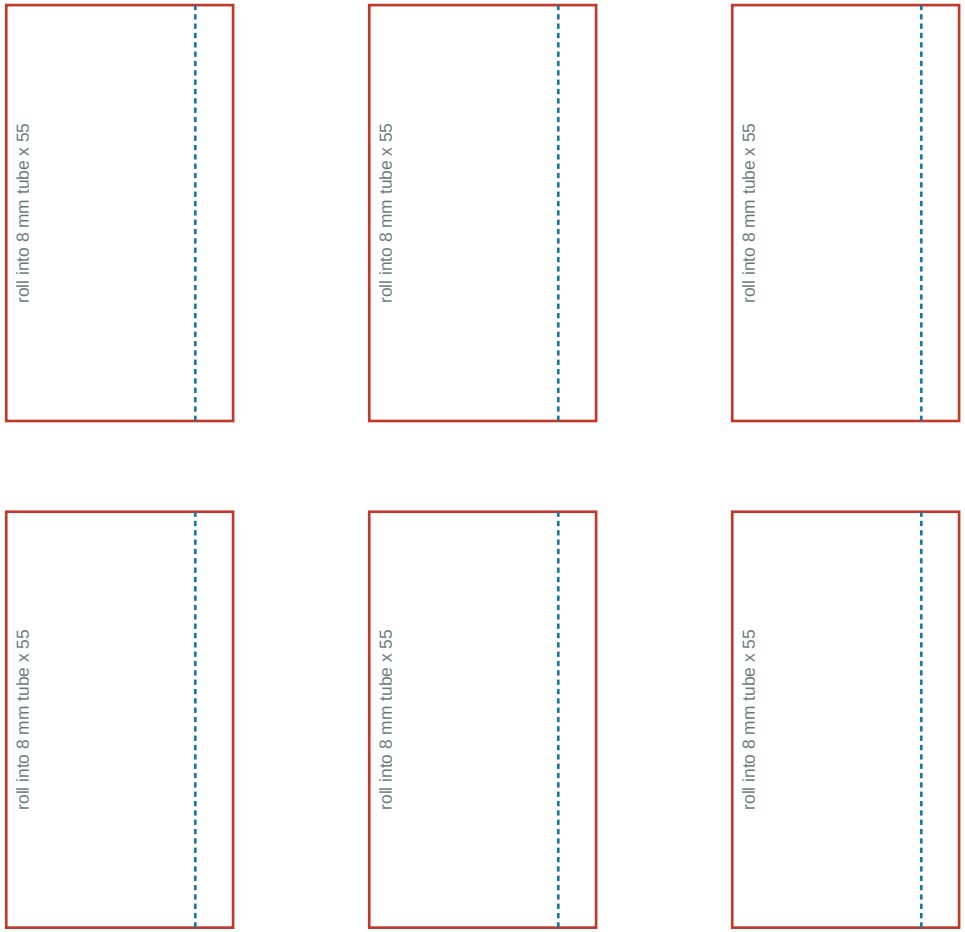


Red solid = cut; blue dashed = fold/roll. Mast templates are visual mockups, not load-bearing parts.

CARDBOARD MOCKUP - SIX SPACER TUBES

Cut six equal strips, roll each to an 8 mm tube, and tape the 5 mm overlap flap.

55 mm stand-offs between the two decks



Each strip is 30 x 55 mm: about 25 mm circumference + 5 mm overlap flap.

Make six. Tape each seam; trim the ends square so both plates sit level.

50 mm CHECK

Do not use cardboard spacer tubes to support powered electronics or a moving robot.